EMPLOYEE SALARY DEMO

Aim

To write a Java program using an interface to display salary details for regular and contract employees.

Description

The Employee interface declares the showSalary() method. RegularEmployee and ContractEmployee implement this interface and calculate salary using their respective pay components. The program reads an employee ID: IDs beginning with R select a regular employee, while IDs beginning with C select a contract employee. It then displays the salary details; other IDs produce an invalid-ID message.

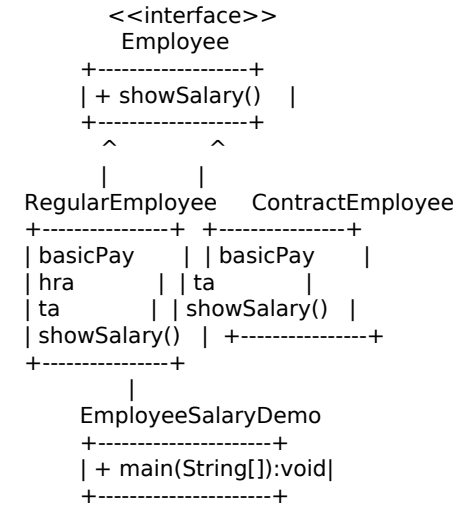
Software Requirements

Hardware: Computer or laptop. Software: Java Development Kit (JDK) and a Java IDE or text editor such as Notepad, VS Code, or Eclipse.

Algorithm

1. Start the program.
2. Create a Scanner object and read the employee ID.
3. Remove spaces at the beginning and end of the ID.
4. If the ID starts with R, create a RegularEmployee object and call showSalary().
5. Otherwise, if the ID starts with C, create a ContractEmployee object and call showSalary().
6. If neither condition is true, display Invalid Employee ID!.
7. Close the Scanner and stop.

UML Diagram



Code

```
package recordprograms;

import java.util.Scanner;

// Interface
interface Employee {
    void showSalary();
}

// Regular Employee class
class RegularEmployee implements Employee {

    int basicPay = 25000;
    int hra = 15000;
    int ta = 5000;

    @Override
    public void showSalary() {
        int total = basicPay + hra + ta;

        System.out.println("Employee Type: Regular");
        System.out.println("Salary Details:");
        System.out.println("Basic Pay: " + basicPay);
        System.out.println("HRA: " + hra);
        System.out.println("TA: " + ta);
        System.out.println("Total Salary: " + total);
    }
}

// Contract Employee class
class ContractEmployee implements Employee {

    int basicPay = 12000;
    int ta = 3000;

    @Override
    public void showSalary() {
        int total = basicPay + ta;

        System.out.println("Employee Type: Contract");
        System.out.println("Salary Details:");
        System.out.println("Basic Pay: " + basicPay);
        System.out.println("HRA: 0");
        System.out.println("TA: " + ta);
        System.out.println("Total Salary: " + total);
    }
}

// Main class
public class EmployeeSalaryDemo {

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Employee ID: ");
        String id = sc.nextLine().trim();

        Employee emp;

        if (id.toUpperCase().startsWith("R")) {
            emp = new RegularEmployee();
            emp.showSalary();
        }
        else if (id.toUpperCase().startsWith("C")) {
            emp = new ContractEmployee();
            emp.showSalary();
        }
        else {
            System.out.println("Invalid Employee ID!");
        }

        sc.close();
```

```
    }
}
```

Output Screenshot

Example 1: Regular employee

```
Enter Employee ID: R101
Employee Type: Regular
Salary Details:
Basic Pay: 25000
HRA: 15000
TA: 5000
Total Salary: 45000
```

Example 2: Contract employee

```
Enter Employee ID: C201
Employee Type: Contract
Salary Details:
Basic Pay: 12000
HRA: 0
TA: 3000
Total Salary: 15000
```

Result

Thus, the Java program using an interface to display salary details for regular and contract employees was written and executed successfully.